

Section A – Conditionals and Simple Logic

Problem 1: Evaluate Glucose Level

Background: A patient's fasting blood glucose (FBG) was measured at **112 mg/dL**.

Task:

Use an if-else statement to classify the glucose value as:

- "Normal" if < 100
- "Prediabetes" if between 100–125
- "Diabetes" if ≥ 126

Problem 2: Check for Fever

Background: A patient's temperature reading is **38.2°C**.

Task:

Write a conditional statement that prints "Fever Detected" if the temperature > 37.5 , otherwise "Normal Temperature".

Problem 3: BMI Classification

Background: You calculated BMI = **29.5** for a study participant.

Task:

Use if-else to classify as "Normal", "Overweight", or "Obese".

Problem 4: Lab Result Flagging

Background: Hemoglobin level is **10.5 g/dL**.

Task:

If $Hb < 12 \rightarrow$ print "Anemia"; otherwise $\rightarrow$ "Normal".

Problem 5: Evaluate Cholesterol Panel

Background: A participant's LDL cholesterol is **165 mg/dL**.

Task:

Write an if-else that prints:

- "Optimal" if < 130
- "Borderline" if 130–159
- "High" if ≥ 160

Section B – Conditionals with Data Frames

Problem 6: Create a Clinical Dataset

Background: You are analyzing 4 patients with the following data:

Name Age BMI BP_Systolic

Aisha 45 31.2 150

Rahman 52 28.5 165

Rima 37 24.1 120

Hossain 60 33.4 175

Task:

Create a data frame named patients.

Problem 7: Add a Hypertension Column

Background: Blood pressure ≥ 140 mmHg indicates hypertension.

Task:

Add a new column "Hypertensive" using `ifelse()` with "Yes" or "No".

Problem 8: Add an Obesity Status Column

Background:

- "Obese" if BMI ≥ 30
- "Overweight" if BMI 25–29.9
- "Normal" if BMI < 25

Task:

Use nested `ifelse()` to classify patients and store in a new column "BMI_Status".

Problem 9: Identify Elderly Patients

Background: A geriatric trial defines elderly as **Age ≥ 50** .

Task:

Filter the data frame to show only elderly participants.

Problem 10: Assign Risk Level

Background:

A hospital uses a simple rule to label cardiac risk:

- High Risk: Age > 50 AND Hypertensive == "Yes"
- Moderate Risk: Only one of these conditions
- Low Risk: None

Task:

Add a new column "Risk_Level" using conditional logic.

Section C – Loops and Iteration

Problem 11: Daily Data Entry Reminder

Background: You want the system to remind lab technicians to upload data each morning for 7 days.

Task:

Use a for loop to print "Please upload lab data for Day X" where X = 1–7.

Problem 12: Sequential Year Tracker

Background: The lab wants a yearly log for 2015 to 2025.

Task:

Use a for loop to print "Processing year: <year>" for each year in that range.

Problem 13: Viral Load Tracking

Background: You have a vector of viral loads (copies/mL): `c(100, 550, 1200, 20000, 850)`.

Task:

Use a for loop and conditional logic:

- Print "Normal" if < 1000
- "Elevated" if between 1000–9999
- "Critical" if ≥ 10000

Section D – Medium to Hard

Problem 14: Automated Clinical Categorization

Background: You have a dataset of HIV patients and their CD4 cell counts.

Name CD4_Count

Rahim 120

Sumaiya 480

Babul 230

Joya 700

CD4 thresholds:

- "Severe Immunodeficiency" if < 200
- "Moderate" if 200–500
- "Normal" if > 500

Task:

1. Create the data frame.
2. Add a new column "Immunity_Status" using ifelse() logic.
3. Print only "Severe" cases.

Problem 15: Integrating Loops and Conditionals for Clinical Scoring

Background: You are developing a simple **Infection Severity Scoring** model for 5 patients.

Each has a numeric infection score stored in a vector: c(10, 35, 50, 80, 95).

Severity classification:

- $< 30 \rightarrow$ "Mild"
- 30–70 $\rightarrow$ "Moderate"
- 70 $\rightarrow$ "Severe"

Task:

1. Use a for loop and if-else to assign severity categories for each patient.
2. Store results in a new vector "Severity_Label".
3. Combine the original scores and labels in a data frame.

Problem 16: Combine All Skills – Hospital Ward Analysis

Background: A hospital ward recorded 6 patients with their age, temperature, and oxygen saturation (SpO₂).

Name Age Temp SpO2

Mina 25 36.8 99

Rafi 50 38.5 95

Sima 61 39.2 91

Rony 45 37.0 98

Asha 70 38.0 89

Name Age Temp SpO₂

Nayeem 58 36.5 97

Clinical Rule:

- If Temp > 38 **AND** SpO₂ < 94 → "Critical"
- Else if Temp > 37 **AND** SpO₂ < 96 → "At Risk"
- Else → "Stable"

Task:

1. Create the data frame.
2. Write a loop or ifelse() structure to classify each patient's condition.
3. Add a column "Condition" with the category.
4. Print a summary table of how many patients fall into each category.